

NVIDIA Corporation

Strong Results/Guide on AI/Accelerated Compute;
Demand Continues to Outstrip Supply Through CY24;
Reit OW

NVIDIA reported Jan results (revenue/GMs/EPS) well above consensus and off of a higher Jan-Qtr revenue base, the team guided to a 9% Q/Q increase in the Apr-Qtr, well-above consensus on continued strong spending from its customers to support their AI/accelerated compute initiatives (both training/inference). More importantly, the team noted that demand will continue to outstrip supply through CY24, allaying fears of an inventory build/correction in the 2H of the year. The team feels constructive on the sustained growth into CY25+ driven by transition to accelerated computing, proliferation of generative AI applications, and build-out of sovereign AI. The team provided examples of the strong AI monetization from their customers (e.g. ServiceNow drove their largest ever net-new annual contract value contribution, Microsoft 365 Copilot grew faster in its first two months than the two previous major Microsoft 365 enterprise suite). China sales were MSD% of datacenter revenues in the Jan-Qtr and the team expects to remain at this level for the Apr-Qtr. The team is sampling China compliant products, which we believe could potentially drive revenue upside later in the year. Datacenter grew 28%+ Q/Q in the Jan-Qtr on strong compute demand (up 27% Q/Q) which is also driving demand for its InfiniBand/Ethernet networking solutions (Networking up 28% Q/Q). In fact, the networking business now exceeds \$13B on an annualized revenue run-rate (from >\$10B in the Oct-Qtr). We estimate the team can drive 10-15% Q/Q growth in the Data Center business for the Apr-Qtr. On the software side, the team exit the Jan-Qtr at a \$1B annualized revenue run-rate (from a “few hundred million” a few quarters ago). Multiple new products across its Datacenter portfolio are also ramping (e.g. H200 and Spectrum-X) to augment the strong forward growth profile. Longer term, the team continues to maintain a 1-2 step lead ahead of competitors with its silicon/hardware/software platforms, and a strong ecosystem (4M CUDA software developers) and the team is further distancing itself with its aggressive cadence of new product launches and more product segmentation over time. We increase our forward estimates and increase our Dec-24 PT to \$850 (from our prior PT of \$650), and reiterate our OW rating.

- **Strong F4Q24 (Jan-Qtr) results led by continued strong AI demand pull for its Data Center products.** NVIDIA reported F4Q24 revenue of \$22.10B (up 22% Q/Q), well above consensus (\$20.4B) led by the continued strong demand pull for its Datacenter solutions. Datacenter revenue grew 27% Q/Q on strong demand pull for its H100 products (DataCenter Compute up 27% Q/Q) and high performance networking products (up 28% Q/Q). In fact, its networking revenues exceeded \$13B on a annualized revenue run-rate. Gaming revenues was flattish Q/Q on seasonal strength and strong demand for its RTX 40 series. Pro Vis (up 11% Q/Q) continues to recover following channel inventory normalization. Auto was up 8% Q/Q. GM of 76.7% was above consensus. EPS of \$5.16 was well-above consensus.

Sources for: Style Exposure – J.P. Morgan Quantitative and Derivatives Strategy; all other tables are company data and J.P. Morgan estimates.

See page 10 for analyst certification and important disclosures.

J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Overweight

NVDA, NVDA US
Price (21 Feb 24):\$674.72

▲ Price Target (Dec-24):\$850.00
Prior (Dec-24):\$650.00

Semiconductors & Semiconductor Capital Equipment / IT Hardware

Harlan Sur AC
(1-415) 315-6700
harlan.sur@jpmorgan.com
Bloomberg JPMA SUR <GO>

Peter K Peng
(1-415) 315-8327
peter.k.peng@jpmchase.com

Pritesh Patel
(1-415) 315-3933
pritesh.n.patel@jpmorgan.com
J.P. Morgan Securities LLC

Key Changes (FYE Jan)

	Prev	Cur
Adj. EPS - 25E (\$)	18.85	22.70

Quarterly Forecasts (FYE Jan)

Adj. EPS (\$)	2024A	2025E	2026E
Q1	1.09	5.32	
Q2	2.70	5.49	
Q3	4.02	5.80	
Q4	5.16	6.09	
FY	12.96	22.70	

Style Exposure

Quant Factors	Current	Hist %Rank (1=Top)			
	%Rank	6M	1Y	3Y	5Y
Value	81	88	84	77	79
Growth	25	80	71	74	33
Momentum	5	1	46	55	93
Quality	2	7	6	7	2
Low Vol	58	63	63	45	72
ESGQ	21	9	11	2	1

Price Performance



	YTD	1m	3m	12m
Abs	36.2%	13.4%	35.1%	226.7%
Rel	31.8%	10.5%	25.3%	202.0%

Company Data

Shares O/S (mn)	2,499
52-week range (\$)	746.11-204.21
Market cap (\$ mn)	1,686,125.00
Exchange rate	1.00
Free float(%)	96.0%
3M - Avg daily vol (mn)	45.00
3M - Avg daily val (\$ mn)	25,324.8
Volatility (90 Day)	37
Index	S&P 500
BBG BUY HOLD SELL	60 5 1

Key Metrics (FYE Jan)

\$ in millions	FY24A	FY25E
Financial Estimates		
Revenue	60,922	104,142
Adj. EBIT	33,069	63,685
Adj. EBITDA	33,493	64,109
Adj. net income	32,312	56,606
Adj. EPS	12.96	22.70
BBG EPS	21.59	25.05
Cashflow from operations	28,091	48,207
FCFF	27,023	46,907
Margins and Growth		
Revenue Growth Y/Y (%)	125.9%	70.9%
EBIT margin	54.3%	61.2%
EBIT growth	464.5%	92.6%
EBITDA margin	55.0%	61.6%
EBITDA Growth Y/Y (%)	433.2%	91.4%
Net margin	53.0%	54.4%
Adj. EPS growth	288.4%	75.2%
Ratios		
Adj. tax rate	12.1%	17.0%
Interest cover	-	-
Net debt/Equity	NM	NM
Net debt/EBITDA	NM	NM
ROE	99.3%	80.0%
Valuation		
FCFF yield	1.6%	2.8%
Dividend yield	0.1%	0.1%
EV/Revenue	27.4	15.6
EV/EBITDA	49.9	25.3
Adj. P/E	52.1	29.7

Summary Investment Thesis and Valuation

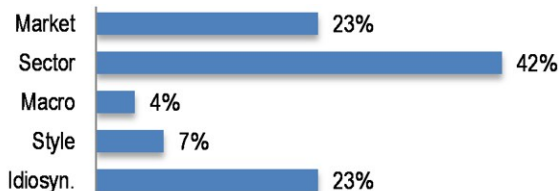
Investment Thesis

We believe NVIDIA continues to execute across all segments. While 1H is typically seasonally weaker than 2H, we expect solid demand in PC gaming to be a strong revenue driver for the company, offsetting PC OEM, which is in secular decline. We expect the data center segment to grow strongly as hyperscale customers continue to embrace GPU-accelerated deep learning for processing large data sets. We are encouraged by strength in the automotive and enterprise segments as well, although strong adoption of autonomous driving in the market remains to be seen. We anticipate significant upside in the shares, driving our Overweight rating.

Valuation

Our Dec-24 PT assumes that NVIDIA trades at 30-35x applied to ~\$29.00 of earnings power in CY25, discounted back to CY24 assuming a 12% discount rate. Our outlook of \$29.00 of earnings power assumes the team will drive its earnings power by 30%+ per year over the next few years led by the continued strong datacenter growth (30-40% CAGR), monetizing an incremental ~\$14B of auto revenue pipeline and an incremental \$2-3B of software, licensing, and subscription revenues over the next 3-4 years.

Performance Drivers



Factors	6M Corr	1Y Corr
Market: MSCI US	0.65	0.48
Sect: Technology	0.78	0.78
Ind: Semicond & S Equip	0.94	0.89
Macro:		
Economic Surprise	0.01	0.25
US 10yr Breakeven	-0.40	-0.21
Crude Oil	-0.23	-0.20
Quant Styles:		
DivYld	-0.41	-0.38
LowVol	-0.08	-0.35
Growth	0.01	0.33

Source: J.P. Morgan Quantitative and Derivatives Strategy for Performance Drivers; company data, Bloomberg Finance L.P. and J.P. Morgan estimates for all other tables. Note: Price history may not be complete or exact.

This document is being provided for the exclusive use of JIAXIN LIU at THE HK UNIVERSITY OF SCI & TECH GZ.

- **F1Q25 (Apr-Qtr) guidance well above consensus on continued strong Datacenter demand driven by accelerated compute/generative AI.** Apr-Q revenue guidance of \$24.0B (up 9% Q/Q, up 234% Y/Y) was well above consensus (\$21.90B) on continued strong DataCenter demand given the strong demand pull from Generative AI/LLMs and on the continued ramp of its Hopper architecture. New products like the H200 and Spectrum-X (ethernet) are beginning to ramp. At the midpoint, in F1Q25, the team sees Q/Q growth led by datacenter. GM is expected to increase to 77.0% on favorable cost but should normalize to 75% in the Jul-Qtr. OpEx is guided to \$2.5B. Implied EPS is \$5.32, well-above consensus of \$4.93.
- **Datacenter revenue continues to accelerate in CY24; demand expected to outstrip supply through CY24.** Demand was strong for AI inference and training. The team estimates 40% of datacenter revenues in CY23 was for AI inference. The team continues to see strong growth in the Apr-Qtr (we estimate 10-15% Q/Q) and beyond. More importantly, the team noted that demand will continue to outpace supply growth through CY24, allaying fears of inventory build/correction in the 2H of the year. The team feels constructive on the sustained growth into CY25+ driven by transition to accelerated computing, proliferation of generative AI applications, and build-out of sovereign AI.
- **Increasing forward estimates and increasing our Dec-24 PT to \$850.** We increase our forward estimates and increase our PT to \$850 (from our prior PT of \$650). Our Dec-24 PT assumes that NVIDIA trades at 30-35x applied to ~\$29.00 of earnings power in CY25, discounted back to CY24 assuming a 12% discount rate. Our outlook of \$29.00 of earnings power assumes the team will drive its earnings power by 30%+ per year over the next few years led by the continued strong datacenter growth (30-40% CAGR), monetizing an incremental ~\$14B of auto revenue pipeline and an incremental \$2-3B of software, licensing, and subscription revenues over the next 3-4 years. With our view of significant upside potential in shares, we are Overweight NVDA.

Table 1: NVIDIA F4Q24 (Jan-Qtr) Earnings and F1Q25 (Apr-Qtr) Outlook

\$ in millions, except per share data

	F4Q24 (Jan-Q)				F1Q25E (Apr-Q)	
	Actual	JPM Est	Diff	Consensus	Guidance	Consensus
Revenue (\$M)	\$22,103	\$20,000	\$2,103	\$20,412	\$24,000	\$21,897
Q/Q Change	22.0%	10.4%	11.6%			
Gross Margin (excl SBC)	76.7%	75.5%	1.2%	75.4%	77.0%	75.5%
Op Margin (excl SBC)	66.2%	63.7%	2.5%			
Proforma EPS (excl SBC)	\$5.16	\$4.33	\$0.83	\$4.60	\$5.32	\$4.93

Source: Company reports and J.P. Morgan estimates

Table 2: Reporting Segment Summary for F4Q24 (vs. F3Q24)

\$ in millions

	F3Q24	F4Q24	Q/Q	Y/Y	% of Revs
Graphic	\$3,475	\$4,205	21%	77%	19%
Compute & Networking	\$14,645	\$17,898	22%	387%	81%
Total	\$18,120	\$ 22,103	22%	265%	

Source: Company reports

Table 3: End Market Segment Summary for F4Q24 (vs. F3Q24)

\$ in millions

	F3Q24	F4Q24	Q/Q	Y/Y	% of Revs
PC OEM	\$73	\$90	23%	7%	0%
PC Gaming	\$2,856	\$2,865	0%	56%	13%
Pro Vis	\$416	\$463	11%	105%	2%
Datacenter	\$14,514	\$18,404	27%	409%	83%
Automotive	\$261	\$281	8%	-4%	1%
Total	\$18,120	\$22,103	22%	265%	

Source: Company reports

Balance Sheet and Cash Flow

During the Jan-Qtr, the company generated cash flow from operations of \$11.5B (vs. \$7.06B in the Oct-Qtr) and exited the quarter with \$25.9B in cash, cash equivalents, and marketable securities (vs. \$18.3B in the Oct-Qtr). Inventory increased to \$5.3B in the Jan-Qtr. In F4Q24, NVIDIA returned \$99 million in cash dividend and repurchased \$2.7B in shares.

Appendix I: Comp and Group Valuation

Table 4: Semiconductor Comp and Group Valuation Table

JPM Rating	Market Cap	2/21/24 Price	Non-GAAP EPS		P/E		Revenue		P/S		Non-GAAP EPS		Sales		P/E		P/S		
			C23E	C24E	C23E	C24E	C23E	C24E	C23E	C24E	C23E	C24E	C23E	C24E	C23E	C24E	C23E	C24E	
Harlan Sur, Lead Coverage phone: 415-315-6700, email: harlan.sur@jpmchase.com																			
PC Semiconductors																			
INTC	UW	\$185,182	\$43.47	\$1.05	\$1.32	41.4	32.8	\$54,228	\$57,853	3.4	3.2	\$1.05	\$1.33	\$54,228	\$57,748	41.4	32.7	3.4	3.2
NVDA	OW	\$1,680,053	\$674.72	\$12.96	\$22.70	52.1	29.7	\$60,922	#####	27.6	16.1	\$11.54	\$22.00	\$55,571	\$96,500	58.5	30.7	30.2	17.4
AMD	N	\$267,464	\$164.29	\$2.65	\$3.50	62.0	46.9	\$22,680	\$25,556	11.8	10.5	\$2.65	\$3.59	\$22,680	\$25,803	62.0	45.8	11.8	10.4
Memory																			
MU	OW	\$89,639	\$81.49	N/A	\$1.88	N/A	43.4	\$16,181	\$25,211	5.5	3.6	(\$4.81)	\$2.53	\$16,724	\$26,460	(16.9)	32.2	5.4	3.4
WDC	NR	\$17,352	\$53.39	N/A	\$3.24	N/A	16.5	\$11,257	\$14,753	1.5	1.2	(\$5.80)	\$3.71	\$11,257	\$15,088	(9.2)	14.4	1.5	1.2
Enterprise/Networking/Datacenter Semiconductors																			
MRVL	OW	\$56,518	\$65.52	\$1.51	\$2.10	43.4	31.2	\$5,501	\$6,197	10.3	9.1	\$1.51	\$1.97	\$5,500	\$5,955	43.4	33.3	10.3	9.5
AVGO	OW	\$524,121	\$1,227.45	\$41.52	\$47.90	29.6	25.6	\$38,362	\$51,924	13.7	10.1	\$42.31	\$49.14	\$37,598	\$50,093	29.0	25.0	13.9	10.5
Mobile Devices																			
SWKS	N	\$16,667	\$103.52	\$7.92	\$7.47	13.1	13.9	\$4,645	\$4,619	3.6	3.6	\$7.92	\$7.11	\$4,645	\$4,516	13.1	14.6	3.6	3.7
QRVO	UW	\$10,870	\$111.89	\$5.09	\$7.79	22.0	14.4	\$3,461	\$4,087	3.1	2.7	\$5.09	\$7.13	\$3,461	\$4,058	22.0	15.7	3.1	2.7
IoT																			
SLAB	OW	\$4,316	\$135.52	\$1.65	N/A	82.1	N/A	\$782	\$660	5.5	6.5	\$1.70	(\$0.43)	\$782	\$683	79.9	N/A	5.5	6.3
SYNA	OW	\$4,105	\$104.73	\$3.47	\$2.67	30.1	39.2	\$1,029	\$1,015	4.0	4.0	\$3.47	\$2.96	\$1,029	\$1,034	30.2	35.3	4.0	4.0
Analog/Microcontrollers																			
TXN	OW	\$150,133	\$164.08	\$7.07	\$5.45	23.2	30.1	\$17,519	\$16,071	8.6	9.3	\$7.07	\$5.37	\$17,519	\$15,738	23.2	30.6	8.6	9.5
ADI	OW	\$96,616	\$193.72	\$9.06	\$6.22	21.4	31.1	\$11,569	\$9,193	8.4	10.5	\$9.40	\$6.08	\$11,814	\$9,217	20.6	31.9	8.2	10.5
NXPI	N	\$61,253	\$235.32	\$14.03	\$13.85	16.8	17.0	\$13,276	\$13,268	4.6	4.6	\$14.01	\$13.54	\$13,276	\$13,133	16.8	17.4	4.6	4.7
MCHP	OW	\$45,622	\$83.48	\$5.98	\$2.56	13.9	32.6	\$8,541	\$5,543	5.3	8.2	\$5.98	\$2.65	\$8,541	\$5,666	14.0	31.5	5.3	8.1
Diversified/Consumer/Standard Components/Other																			
ON	N	\$34,114	\$77.62	\$5.15	\$4.33	15.1	17.9	\$8,253	\$7,592	4.1	4.5	\$5.16	\$4.30	\$8,253	\$7,534	15.0	18.1	4.1	4.5
MTSI	N	\$6,126	\$84.74	\$2.47	\$2.95	34.3	28.7	\$625	\$768	9.8	8.0	\$2.47	\$2.71	\$625	\$768	34.3	31.3	9.8	8.0
Foundries																			
GFS	N	\$29,738	\$53.39	\$2.24	\$1.40	23.8	38.1	\$7,392	\$6,885	4.0	4.3	\$2.24	\$1.35	\$7,392	\$6,793	23.8	39.6	4.0	4.4
Semiconductor Capital Equipment																			
AMAT	OW	\$159,306	\$190.33	\$8.15	\$8.94	23.4	21.3	\$26,485	\$28,100	6.0	5.7	\$8.12	\$8.33	\$26,498	\$27,049	23.4	22.9	6.0	5.9
LRGX	OW	\$119,253	\$901.93	\$27.34	\$31.47	33.0	28.7	\$14,317	\$15,597	8.3	7.6	\$27.34	\$31.17	\$14,317	\$15,804	33.0	28.9	8.3	7.5
KLAC	OW	\$88,827	\$651.92	\$22.78	\$24.98	28.6	26.1	\$9,671	\$10,150	9.2	8.8	\$22.79	\$24.31	\$9,671	\$10,084	28.6	26.8	9.2	8.8
Chip Design Automation Software																			
ARM	OW	\$129,422	\$123.39	\$0.91	\$1.25	135.2	98.4	\$2,938	\$3,523	44.1	36.7	\$1.06	\$1.31	\$3,056	\$3,655	116.8	94.2	42.4	35.4
CDNS	N	\$79,209	\$290.76	\$5.15	\$5.92	56.5	49.1	\$4,090	\$4,580	19.4	17.3	\$5.15	\$5.93	\$4,090	\$4,588	56.5	49.0	19.4	17.3
SNPS	OW	\$84,435	\$543.57	\$12.01	\$13.67	45.3	39.8	\$6,126	\$6,823	13.8	12.4	\$11.84	\$13.59	\$6,035	\$6,710	45.9	40.0	14.0	12.6
Large Cap Semi Average						31.7	29.1			9.4	8.2					32.2	29.3	9.7	8.4
Small-Mid Cap Semi Average						38.4	20.3			5.6	5.4					23.8	21.7	5.7	5.1
Semi Group Average (ex-outliers)						24.8	26.1			6.6	6.5					24.7	26.8	6.6	6.6
SOX Index			4,446	114	164	39.2	27.1			6.8	6.4					39.2	27.1	6.8	6.4
S&P500			4,982	240	270	20.7	18.5			2.7	2.5					20.7	18.5	2.7	2.5

Source: Company reports, Bloomberg Finance L.P., and J.P. Morgan estimates. Prices as of market close 02/21/2024

Appendix II: Financial Tables

Table 5: NVIDIA Income Statement

\$ in millions, except per share data

\$M	Apr-21	Jul-21	Oct-21	Jan-22	C21	Apr-22	Jul-22	Oct-22	Jan-23	C22	Apr-23	Jul-23	Oct-23	Jan-24	C23	Apr-24	Jul-24	Oct-24	Jan-25	C24
FY end: Jan.	1Q22	2Q22	3Q22	4Q22	F22	1Q23	2Q23	3Q23	4Q23	F23	1Q24	2Q24	3Q24	4Q24	F24	1Q25E	2Q25E	3Q25E	4Q25E	F25E
Total revenue	5,661	6,507	7,103	7,643	26,914	8,288	6,704	5,931	6,051	26,974	7,192	13,507	18,120	22,103	60,922	24,000	25,461	26,693	27,988	104,142
% Change Y/Y	83.8%	68.3%	50.3%	52.8%	61.4%	46.4%	3.0%	-16.5%	-20.8%	0.2%	-13.2%	101.5%	205.5%	265.3%	125.9%	233.7%	88.5%	47.3%	26.6%	70.9%
% Change Q/Q	13.2%	14.9%	9.2%	7.6%		8.4%	-19.1%	-11.5%	2.0%		18.9%	87.8%	34.2%	22.0%		8.6%	6.1%	4.8%	4.8%	
Cost of goods sold	2,032	2,292	2,472	2,644	9,440	2,857	3,789	2,754	2,218	11,618	2,544	4,045	4,720	5,312	16,621	5,688	6,543	6,780	7,081	26,093
% Change Y/Y	88.8%	44.1%	40.0%	43.2%	50.3%	40.6%	65.3%	11.4%	-16.1%	23.1%	-11.0%	6.8%	71.4%	139.5%	43.1%	123.6%	61.8%	43.6%	33.3%	57.0%
% Change Q/Q	10.1%	12.8%	7.9%	7.0%		8.1%	32.6%	-27.3%	-19.5%		14.7%	59.0%	16.7%	12.5%		7.1%	15.0%	3.6%	4.4%	
Non-GAAP Gross margin (ex SBC)	66.2%	66.7%	67.0%	67.0%	66.8%	67.1%	45.9%	56.1%	66.1%	57.4%	66.8%	71.2%	75.0%	76.7%	72.9%	77.0%	75.0%	75.2%	75.3%	75.1%
GAAP Gross margin	64.1%	64.8%	65.2%	65.4%	64.9%	65.5%	43.5%	53.6%	63.3%	56.9%	64.6%	70.1%	74.0%	76.0%	72.7%	76.3%	74.3%	74.6%	74.7%	74.9%
R&D	1,153	1,245	1,403	1,466	5,267	1,618	1,824	1,945	1,951	7,338	1,875	2,040	2,294	2,465	8,674	2,748	2,803	2,860	2,917	11,328
% Total revenue	20.4%	19.1%	19.8%	19.2%	19.6%	19.5%	27.2%	32.8%	32.2%	27.2%	26.1%	15.1%	12.7%	11.2%	14.2%	11.5%	11.0%	10.7%	10.4%	10.9%
SG&A (Adj.)	440	454	487	493	1,874	530	536	561	533	2,160	583	609	673	693	2,558	731	746	761	799	3,036
% Total revenue	7.8%	7.0%	6.9%	6.5%	7.0%	6.4%	8.0%	9.5%	8.8%	8.0%	8.1%	4.5%	3.7%	3.1%	4.2%	3.0%	2.9%	2.8%	2.9%	2.9%
Operating income	2,036	2,516	2,741	3,040	10,333	3,283	555	671	1,349	5,858	2,190	6,813	10,433	13,633	33,069	14,833	15,368	16,293	17,191	63,685
% Total revenue	36.0%	38.7%	38.6%	39.8%	38.4%	39.6%	8.3%	11.3%	22.3%	21.7%	30.5%	50.4%	57.6%	61.7%	54.3%	61.8%	60.4%	61.0%	61.4%	61.2%
Opex GAAP	1673	1771	1960	2029		3563	2416	2576	2576		2508	2662	2983	3176		3480	3549	3620	3715	
Opex Non-GAAP	1189	1266	1375	1447	5,277	1608	1749	1793	1775	6,925	1750	1838	2026	2210	7,824	2532	2601	2672	2767	10,572
Opm % ex options	43.5%	45.8%	46.5%	47.0%	47.2%	46.6%	18.0%	23.9%	34.5%	31.8%	40.7%	56.7%	63.0%	66.2%	60.1%	65.9%	64.3%	64.8%	65.0%	65.0%
Interest/Other Expense	(45)	(49)	(52)	(52)	(198)	(45)	(16)	24	43	6	84	120	175	232	611	250	250	250	250	1,000
Pretax income	1,991	2,467	2,689	2,988	10,135	3,238	539	695	1,392	5,864	2,274	6,933	10,608	13,865	33,680	15,083	15,618	16,543	17,441	64,685
% Total revenue	35.2%	37.9%	37.9%	39.1%	37.7%	39.1%	8.0%	11.7%	23.0%	21.7%	31.6%	51.3%	58.5%	62.7%	55.3%	62.8%	61.3%	62.0%	62.3%	62.1%
Income tax	132	20	174	(138)	188	187	(181)	(67)	(125)	(186)	166	793	1,279	1,821	4,059	2,564	2,655	2,812	2,965	10,997
Tax rate	6.6%	0.8%	6.5%	-4.6%	1.9%	5.8%	-33.6%	-9.6%	-9.0%	-3.2%	7.3%	11.4%	12.1%	13.1%	12.1%	17.0%	17.0%	17.0%	17.0%	17.0%
Net income (non-GAAP)	2,313	2,623	2,973	3,350	11,259	3,443	1,292	1,456	2,174	8,365	2,713	6,740	10,020	12,839	32,312	13,248	13,692	14,460	15,206	56,606
% Total revenue	32.8%	37.6%	35.4%	40.9%	37.0%	36.8%	10.7%	12.8%	25.1%	22.4%	29.3%	45.5%	51.5%	54.5%	48.6%	52.2%	50.9%	51.4%	51.7%	51.6%
Avg. No. of Diluted Shr	2,528.0	2,532.0	2,538.0	2,545.0	2,535.8	2,537.0	2,516.0	2,499.0	2,477.0	2,507.3	2,490.0	2,499.0	2,494.0	2,490.0	2,493.3	2,491.0	2,492.0	2,494.0	2,496.0	2,493.3
Avg. No. of Diluted Shr (non-GAAP)	2,528.0	2,532.0	2,538.0	2,545.0	2,535.8	2,537.0	2,516.0	2,499.0	2,477.0	2,507.3	2,490.0	2,499.0	2,494.0	2,490.0	2,493.3	2,491.0	2,492.0	2,494.0	2,496.0	2,493.3
Convertible Adjustment																				
EPS (Non-GAAP)	\$0.91	\$1.04	\$1.17	\$1.32	\$4.44	\$1.36	\$0.51	\$0.58	\$0.88	\$3.34	\$1.09	\$2.70	\$4.02	\$5.16	\$12.96	\$5.32	\$5.49	\$5.80	\$6.09	\$22.70
% Change Y/Y	103.4%	89.9%	61.0%	69.8%	77.5%	48.3%	-50.4%	-50.3%	-33.3%	-24.9%	-19.7%	425.2%	589.6%	487.5%	288.4%	388.1%	103.7%	44.3%	18.1%	75.2%
% Change Q/Q	18.0%	13.2%	13.1%	12.4%		3.1%	-62.2%	13.5%	50.6%		24.1%	147.5%	49.0%	28.3%		3.1%	3.3%	5.5%	5.1%	

Source: Company reports and J.P. Morgan estimates

Table 6: NVIDIA Balance Sheet

\$ in millions, except per share data

	Jan21	Apr21	Jul21	Oct21	Jan22	Apr22	Jul22	Oct22	Jan23	Apr23	Jul23	Oct23	Jan24	Apr24	Jul24	Oct24	Jan25
FY End: Jan.	4Q21	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25E	2Q25E	3Q25E	4Q25E
Assets																	
Cash and ST investment	11,561	12,667	19,654	19,298	21,208	20,338	17,037	13,143	13,296	15,320	16,023	18,281	25,984	32,053	43,698	57,230	72,495
Accounts receivable	2,429	3,024	3,586	3,954	4,650	5,438	5,317	4,908	3,827	4,080	7,066	8,309	9,999	13,119	13,775	14,552	14,598
Inventories	1,826	1,992	2,114	2,233	2,605	3,163	3,889	4,454	5,159	4,611	4,319	4,779	5,282	9,051	10,051	10,144	9,945
Prepaid and other	239	444	452	321	366	636	1,175	718	791	872	1,389	1,289	3,080	3,080	3,080	3,080	3,080
Total current assets	16,055	18,127	25,806	25,806	28,829	29,575	27,418	23,223	23,073	24,883	28,797	32,658	44,345	57,303	70,604	85,006	100,118
PP&E, net	2,149	2,268	2,364	2,509	2,778	2,916	3,233	3,774	3,807	3,740	3,799	3,844	3,914	4,133	4,352	4,571	4,790
Deposits and other	3,657	3,595	3,809	5,561	5,892	6,145	6,417	7,269	8,254	9,866	11,134	11,965	11,927	11,927	11,927	11,927	11,927
Goodwill	4,193	4,193	4,193	4,302	4,349	4,365	4,372	4,372	4,372	4,430	4,430	4,430	4,430	4,430	4,430	4,430	4,430
Intangible assets	2,737	2,613	2,478	2,454	2,339	2,211	2,036	1,850	1,676	1,541	1,395	1,251	1,112	1,112	1,112	1,112	1,112
Total assets	28,791	30,796	38,650	40,632	44,187	45,212	43,476	40,488	41,182	44,460	49,555	54,148	65,728	78,905	92,425	107,046	122,377
Liabilities																	
Accounts payable	1,201	1,218	1,474	1,664	1,783	1,999	2,421	1,491	1,193	1,141	1,929	2,380	2,699	3,058	3,518	3,645	3,807
Accrued liabilities	1,725	1,787	1,974	1,948	2,552	3,563	3,903	4,115	4,120	4,869	7,156	5,472	6,682	6,682	6,682	6,682	6,682
Short-term Debt	999	999	1,000	0	0	0	1,249	1,249	1,250	1,250	1,249	1,249	1,250	1,250	1,250	1,250	1,250
Deferred revenue																	
Total current liabilities	3,925	4,004	4,448	3,612	4,335	5,562	7,573	6,855	6,563	7,260	10,334	9,101	10,631	10,990	11,450	11,577	11,739
Capital lease obligations	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Long-term Debt	5,964	5,964	10,943	10,944	10,946	10,947	9,700	9,701	9,703	9,704	8,456	8,457	8,459	8,459	8,459	8,459	8,459
Long-term liabilities	2,009	2,054	2,112	2,278	2,294	2,383	2,352	2,583	2,815	2,976	3,264	3,325	3,660	3,660	3,660	3,660	3,660
Total liabilities	11,898	12,022	17,503	16,834	17,575	18,892	19,625	19,139	19,081	19,940	22,054	20,883	22,750	23,109	23,569	23,696	23,858
Total stockholders' equity	16,893	18,774	21,147	23,798	26,612	26,320	23,851	21,349	22,101	24,520	27,501	33,265	42,978	55,796	68,856	83,349	98,520
Total liability and stockholders' equity	28,791	30,796	38,650	40,632	44,187	45,212	43,476	40,488	41,182	44,460	49,555	54,148	65,728	78,905	92,425	107,046	122,377

Source: Company reports and J.P. Morgan estimates

Table 7: NVIDIA Cash Flow Statement

\$ in millions

	Apr'21	Jul'21	Oct'21	Jan'22	Apr'22	Jul'22	Oct'22	Jan'23	Apr'23	Jul'23	Oct'23	Jan'24	Apr'24	Jul'24	Oct'24	Jan'25
FY End: Jan.	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25E	2Q25E	3Q25E	4Q25E
Net income (GAAP)	1,912	2,374	2,464	3,003	1,618	656	680	1,414	2,043	6,188	9,243	12,285	12,519	12,963	13,731	14,476
Depreciation and amortization	281	286	298	309	334	378	406	426	384	365	372	387	106	106	106	106
Stock-based compensation	429	465	559	551	578	648	745	738	735	842	979	993	1,116	1,116	1,116	1,116
Allowance for doubtful accounts																
Excess tax benefits from stock comp																
Other adjustments	(112)	18	31	(151)	851	(5)	(34)	(617)	(1,155)	(69)	3	(447)				
Change in working capital	(595)	(563)	(366)	(692)	(788)	120	(410)	1,081	(252)	(2,986)	(1,243)	(1,690)	(3,120)	(657)	(777)	(46)
Change in Receivables	(159)	(123)	(118)	(374)	(560)	(725)	(563)	(706)	566	296	(457)	(503)	(3,769)	(999)	(93)	199
Prepaid expenses and other assets	2	16	(1,575)	(158)	(1,261)	(293)	247	(210)	(215)	(376)	254	(1,184)	0	0	0	0
Deposits and other assets																
Accounts payable	70	209	15	183	255	304	(917)	(193)	11	777	461	281	(359)	(460)	(127)	(162)
Accrued liabilities and other liabilities	(1)	133	(62)	423	634	633	(92)	166	689	1,986	(1,722)	1,072	0	0	0	0
Other liabilities	47	(133)	273	(61)	70	(446)	330	150	105	(675)	(557)	305				
Cash Flow from operations	1,874	2,682	1,519	3,033	1,731	1,270	392	2,249	2,911	6,348	7,333	11,499	6,493	12,069	13,956	15,690
Purchases of marketable securities	(4,470)	(4,798)	2,756		(3,932)	(3,644)	5,884		(2,801)	(2,542)	2,891					
Sales and maturities of marketable securities	3,498	2,443	(6,752)	(1,243)	6,976	5,738	(2,188)	500	2,512	2,598	(5,345)	(5,743)				
Purchases of PP&E	(298)	(183)	(221)	(273)	(361)	(433)	(530)	(509)	(248)	(289)	(278)	(253)	(325)	(325)	(325)	(325)
Acquisition, net of cash acquired	0	0	(203)	(60)	(36)	(13)	0	0	(83)	0	0	0				
Other Investing Activities	(2)	5	(18)	(11)	(35)	(30)	(18)	5	(221)	(214)	(437)	(113)	(325)	(325)	(325)	(325)
Cash flow from investments	(1,272)	(2,533)	(4,438)	(1,587)	2,612	1,618	3,148	(4)	(841)	(447)	(3,169)	(6,109)	(325)	(325)	(325)	(325)
Common stock issued under employee stock plans	126	2	149		204	1	143		246	1	156		0	0	0	0
Issue/Repurchase of common stock	(477)				(1,996)	(3,345)	(3,485)	(1,212)	0	(3,067)	(3,806)	(2,660)				
Principal payments on capital leases			(1,008)				0				0					
Debt proceeds/borrowings	0	4,985		(26)	0	0		(7)	0	(1,250)		(29)				
Excess tax benefits from stock comp		(365)	(440)	(618)		(305)	(294)	(339)		(672)	(764)	(841)				
Dividends paid	(99)	(100)	(100)	(100)	(100)	(100)	(100)	(98)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)
Other financing activities	(21)	(21)	(22)		(554)	(13)	(17)		(527)	(11)	(14)					
Cash flow from financing	(471)	4,501	(1,421)	(744)	(2,446)	(3,762)	(3,753)	(1,656)	(380)	(5,098)	(4,527)	(3,629)	(99)	(99)	(99)	(99)
Cash at beginning of period	847	978	5,628	1,288	1,990	3,887	3,013	2,800	3,389	5,079	5,882	5,519	7,280	13,349	24,994	38,526
Net change in cash and equivalents	131	4,650	(4,340)	702	1,897	(874)	(213)	589	1,690	803	(363)	1,761	6,069	11,645	13,532	15,266
Cash at end of period	978	5,628	1,288	1,990	3,887	3,013	2,800	3,389	5,079	5,882	5,519	7,280	13,349	24,994	38,526	53,791

Source: Company reports and J.P. Morgan estimate

Investment Thesis, Valuation and Risks

NVIDIA Corporation (*Overweight; Price Target: \$850.00*)

Investment Thesis

We believe NVIDIA continues to execute across all segments. While 1H is typically seasonally weaker than 2H, we expect solid demand in PC gaming to be a strong revenue driver for the company, offsetting PC OEM, which is in secular decline. We expect the data center segment to grow strongly as hyperscale customers continue to embrace GPU-accelerated deep learning for processing large data sets. We are encouraged by strength in the automotive and enterprise segments as well, although strong adoption of autonomous driving in the market remains to be seen. We anticipate significant upside in shares, driving our Overweight rating.

Valuation

Our Dec-24 PT assumes that NVIDIA trades at 30-35x applied to ~\$29.00 of earnings power in CY25E, discounted back to CY24 assuming a 12% discount rate. Our outlook of \$29.00 of earnings power assumes the team will drive its earnings power by 30%+ per year over the next few years led by the continued strong datacenter growth (30-40% CAGR), monetizing an incremental ~\$14B of auto revenue pipeline and an incremental \$1-2B of software, licensing, and subscription revenues over the next 3-4 years.

Risks to Rating and Price Target

- Although PC gaming demand seems resilient to macroeconomic weakness, any macro uncertainty could weigh on PC gaming demand trends. Given NVDA's ~53% exposure to the PC gaming segment, any consumer PC gaming weakness poses downside risk to our estimates.
- NVIDIA's GPUs could gain lower than expected deployment into data center applications as hyperscale customers further adopt deep learning as a new and effective way of processing large unstructured data sets. Any significant decrease in the adoption of deep learning by hyperscale customers or increase in competition could result in downside risk to our revenue and earnings estimates.

NVIDIA Corporation: Summary of Financials

Income Statement - Annual						Income Statement - Quarterly					
	FY23A	FY24A	FY25E	FY26E	FY27E		1Q25E	2Q25E	3Q25E	4Q25E	
Revenue	26,974	60,922	104,142	-	-	Revenue	24,000	25,461	26,693	27,988	
COGS	(11,618)	(16,621)	(26,093)	-	-	COGS	(5,688)	(6,543)	(6,780)	(7,081)	
Gross profit	15,356	44,301	78,050	-	-	Gross profit	18,312	18,917	19,913	20,907	
SG&A	(2,160)	(2,558)	(3,036)	-	-	SG&A	(731)	(746)	(761)	(799)	
Adj. EBITDA	6,282	33,493	64,109	-	-	Adj. EBITDA	14,939	15,474	16,399	17,297	
D&A	(424)	(424)	(424)	-	-	D&A	(106)	(106)	(106)	(106)	
Adj. EBIT	5,858	33,069	63,685	-	-	Adj. EBIT	14,833	15,368	16,293	17,191	
Net Interest	-	-	-	-	-	Net Interest	-	-	-	-	
Adj. PBT	5,864	33,680	64,685	-	-	Adj. PBT	15,083	15,618	16,543	17,441	
Tax	186	(4,059)	(10,997)	-	-	Tax	(2,564)	(2,655)	(2,812)	(2,965)	
Minority Interest	-	-	-	-	-	Minority Interest	-	-	-	-	
Adj. Net Income	8,365	32,312	56,606	-	-	Adj. Net Income	13,248	13,692	14,460	15,206	
Reported EPS	1.74	11.94	21.34	-	-	Reported EPS	4.98	5.15	5.46	5.75	
Adj. EPS	3.34	12.96	22.70	-	-	Adj. EPS	5.32	5.49	5.80	6.09	
DPS	0.68	0.68	0.68	-	-	DPS	0.17	0.17	0.17	0.17	
Payout ratio	39.0%	5.7%	3.2%	-	-	Payout ratio	3.4%	3.3%	3.1%	3.0%	
Shares outstanding	2,507	2,493	2,493	-	-	Shares outstanding	2,491	2,492	2,494	2,496	
Balance Sheet & Cash Flow Statement						Ratio Analysis					
	FY23A	FY24A	FY25E	FY26E	FY27E		FY23A	FY24A	FY25E	FY26E	FY27E
Cash and cash equivalents	13,296	25,984	72,495	-	-	Gross margin	56.9%	72.7%	74.9%	-	-
Accounts receivable	3,827	9,999	14,598	-	-	EBITDA margin	23.3%	55.0%	61.6%	-	-
Inventories	5,159	5,282	9,945	-	-	EBIT margin	21.7%	54.3%	61.2%	-	-
Other current assets	791	3,080	3,080	-	-	Net profit margin	31.0%	53.0%	54.4%	-	-
Current assets	23,073	44,345	100,118	-	-	ROE	34.3%	99.3%	80.0%	-	-
PP&E	3,807	3,914	4,790	-	-	ROA	19.6%	60.4%	60.2%	-	-
LT investments	-	-	-	-	-	ROCE	16.1%	67.8%	65.7%	-	-
Other non current assets	14,302	17,469	17,469	-	-	SG&A/Sales	8.0%	4.2%	2.9%	-	-
Total assets	41,182	65,728	122,377	-	-	Net debt/equity	NM	NM	NM	-	-
Short term borrowings	1,250	1,250	1,250	-	-	P/E (x)	202.2	52.1	29.7	-	-
Payables	1,193	2,699	3,807	-	-	P/BV (x)	76.5	39.1	17.1	-	-
Other short term liabilities	4,120	6,682	6,682	-	-	EV/EBITDA (x)	268.0	49.9	25.3	-	-
Current liabilities	6,563	10,631	11,739	-	-	Dividend Yield	0.1%	0.1%	0.1%	-	-
Long-term debt	9,703	8,459	8,459	-	-	Sales/Assets (x)	0.6	1.1	1.1	-	-
Other long term liabilities	2,815	3,660	3,660	-	-	Interest cover (x)	-	-	-	-	-
Total liabilities	19,081	22,750	23,858	-	-	Operating leverage	(19426.5%)	369.1%	130.5%	-	-
Shareholders' equity	22,101	42,978	98,520	-	-	Revenue y/y Growth	0.2%	125.9%	70.9%	-	-
Minority interests	-	-	-	-	-	EBITDA y/y Growth	(41.6%)	433.2%	91.4%	-	-
Total liabilities & equity	41,182	65,728	122,377	-	-	Tax rate	3.2%	12.1%	17.0%	-	-
BVPS	8.81	17.24	39.51	-	-	Adj. Net Income y/y Growth	(25.7%)	286.3%	75.2%	-	-
y/y Growth	(16.0%)	95.6%	129.2%	-	-	EPS y/y Growth	(24.9%)	288.4%	75.2%	-	-
Net debt/(cash)	(2,343)	(16,275)	(62,786)	-	-	DPS y/y Growth	0.0%	0.0%	0.0%	-	-
Cash flow from operating activities	5,642	28,091	48,207	-	-						
o/w Depreciation & amortization	424	424	424	-	-						
o/w Changes in working capital	(3,174)	(5,057)	(10,370)	-	-						
Cash flow from investing activities	7,374	(10,566)	(1,300)	-	-						
o/w Capital expenditure	(1,833)	(1,068)	(1,300)	-	-						
as % of sales	6.8%	1.8%	1.2%	-	-						
Cash flow from financing activities	(11,617)	(13,634)	(396)	-	-						
o/w Dividends paid	(398)	(396)	(396)	-	-						
o/w Net debt issued/(repaid)	(7)	(1,279)	0	-	-						
Net change in cash	1,399	3,891	46,511	-	-						
Adj. Free cash flow to firm	3,809	27,023	46,907	-	-						
y/y Growth	(53.2%)	609.5%	73.6%	-	-						

Source: Company reports and J.P. Morgan estimates.

Note: \$ in millions (except per-share data). Fiscal year ends Jan. o/w - out of which

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker:** J.P. Morgan Securities LLC makes a market in the securities of NVIDIA Corporation.
- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to NVIDIA Corporation.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: NVIDIA Corporation.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-investment-banking, securities-related: NVIDIA Corporation.
- **Client/Non-Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-securities-related: NVIDIA Corporation.
- **Non-Investment Banking Compensation Received:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from NVIDIA Corporation.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of NVIDIA Corporation, if any.

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables, are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

NVIDIA Corporation (NVDA, NVDA US) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 04, 2002. All share prices are as of market close on the previous business day.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period. J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); and Underweight (over the duration of the price target indicated in

this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, <https://www.jpmorganmarkets.com>.

Coverage Universe: Sur, Harlan: Advanced Micro Devices (AMD), Analog Devices (ADI), Applied Materials (AMAT), Arm Holdings Plc (ARM), Broadcom Inc (AVGO), Cadence Design Systems (CDNS), GlobalFoundries (GFS), Intel (INTC), KLA Corporation (KLAC), Lam Research (LRCX), MACOM (MTSI), Marvell Technology Inc (MRVL), Microchip Technology (MCHP), Micron Technology (MU), NVIDIA Corporation (NVDA), NXP Semiconductors (NXPI), ON Semiconductor Corporation (ON), Qorvo (QRVO), Skyworks Solutions (SWKS), Synopsys Inc (SNPS), Texas Instruments (TXN), Western Digital (WDC)

J.P. Morgan Equity Research Ratings Distribution, as of January 01, 2024

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	47%	39%	13%
IB clients**	48%	43%	32%
JPMS Equity Research Coverage*	46%	42%	12%
IB clients**	68%	63%	46%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or http://www.finra.org/sites/default/files/Security_Futures_Risk_Disclosure_Statement_2018.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below, this material has been distributed by the legal entity responsible for production. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA" - Central Bank of Argentina) and Comisión Nacional de Valores ("CNV" - Argentinian Securities Commission - ALYC y AN Integral N°51). **Australia:** J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#). **Brazil:** Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmorgan.com. **Canada:** J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P. Morgan Securities Canada Inc. **Chile:** Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile. **China:** J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business. **Dubai International Financial Centre (DIFC):** JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules. **European Economic Area (EEA):** Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons. **Hong Kong:** J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A.,

Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited. **India:** J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmipl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Spurthi Gadamsetty; spurthi.gadamsetty@jpmchase.com; +912261573225. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000.

Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK). **Korea:** J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch. **Japan:** JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan. **Malaysia:** This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia. **Mexico:** J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission. **New Zealand:** This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008. **Philippines:** J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission. **Singapore:** This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MCI (P) 030/08/2023 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material. **South Africa:** J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA). **Taiwan:** J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material. **Thailand:** This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500. **UK:** Unless specified to the contrary, research is distributed in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority. JPMS plc is registered in England & Wales No. 2711006, Registered Office 25 Bank Street, London, E14 5JP. This material is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. Research issued by JPMS plc has been prepared in accordance with JPMS plc's policy for prevention and avoidance of conflicts of interest related to the production of Research which can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#). U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2024. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

"Other Disclosures" last revised February 03, 2024.

Copyright 2024 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party. It is permissible to use J.P. Morgan Data for internal business purposes only in an AI system or model that protects the confidentiality of J.P. Morgan Data so as to prevent any and all access to or use of such J.P. Morgan Data by any third-party.

Completed 22 Feb 2024 01:53 AM EST

Disseminated 22 Feb 2024 01:53 AM EST